

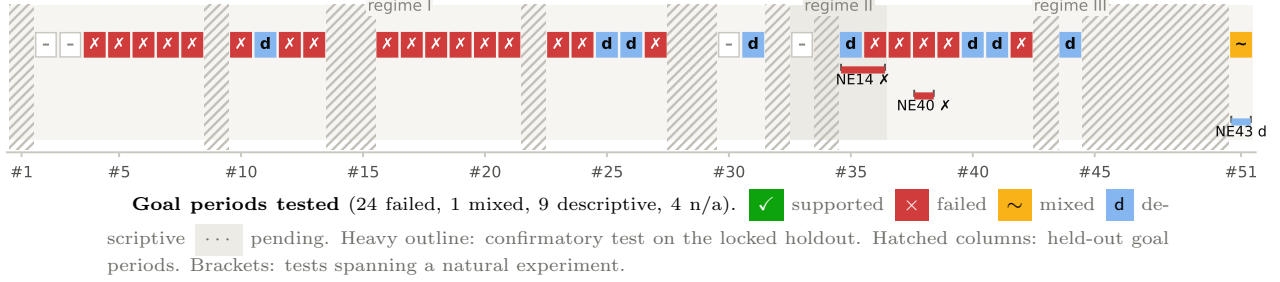
H56: Entropy production fingerprints the platform

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 1.5/5

Bottom line. Action-chain entropy production does not fingerprint the platform: scaffold changes, even NE14b and the undocumented oracle swap, sit inside the work-driven day-to-day spread.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The claim: the entropy production (EP) of agents' turn-level action chains jumps at scaffold changes (prompts, tools, harness), not at goal, roster or room changes. EP would then flag undocumented platform changes. The estimator is a within-agent, count-matched EP change over 3+3 days. It runs at 107 dated events (changelog, NE catalog, kickoffs, joins, rooms) and at every day for blind change-point detection. It runs on all records and on an agent-only chain without scaffold records. Synthetic chains come first: the estimator is calibrated and detects $\pm 50\%$ changes. On real data, scaffold changes do not stand out (0/17 tool-change hits; class $p = 0.23$). Neither does NE14b (-28% , $t = -1.3$), nor the oracle swap (NE40), dated here to 04-20 from answer style ($t = -0.2$). Goal kickoffs move the agent-only chain more ($p = 0.007$). Family differences survive scaffold removal. The holdout is not run.

PHYSICAL MODEL

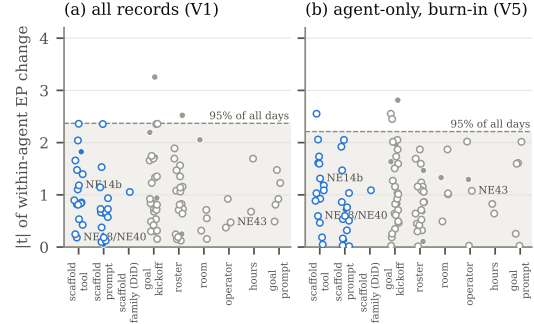
Kinetic Potts (model 02), read through model 05. Each agent is a Markov chain on H14's 11 action classes (or 6 coarse states). Its EP per transition is the pair-KL between forward and reversed transitions. The AIK bound on antisymmetric transition indicators g_{ab} estimates it (Newton step, cross-fitted). H56 treats goal, roster and room changes as fields that move occupancy (potentials U). It treats scaffold changes as changes in the cycle forces F , so only scaffold changes move Σ .

$$\Sigma = \sum_{a,b} \pi_a P_{ab} \ln \frac{\pi_a P_{ab}}{\pi_b P_{ba}}, \quad \ln \frac{P_{ab}}{P_{ba}} = F_{ab} + U_a - U_b$$

$$\hat{\Sigma}_N = \frac{2}{k(k-1)} \sum_{f \neq f'} \bar{g}_f^\top K^{-1} \bar{g}_{f'}, \quad g_{ab} = \mathcal{K}[a \rightarrow b] - \mathcal{K}[b \rightarrow a]$$

$$\Delta_i = \hat{\Sigma}_i(\text{post}) - \hat{\Sigma}_i(\text{pre}) \quad (\text{both sides subsampled to } m_i = \min n)$$

$$t = \bar{\Delta} / \text{SE}(\Delta_i), \quad f_+ = \#\{\Delta_i > 0\} / n$$



Observables. $|t|$ of the within-agent EP change at every eligible event, by class. Filled: unconfounded window. Open: another event in the window. Dashed: 95th percentile over all days. (a) All records. (b) Agent-only chain with a 3-transition burn-in after every scaffold cut. Scaffold changes (blue) sit in the same band as kickoffs, joins and room changes. NE14b (regime II→III) and NE18/NE40 (the oracle swap) are ordinary.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not monitor behavioral EP for silent platform changes. It tracks the work: goal kickoffs move it more than tool changes do, and families differ by tool-use style, not by harness. Monitor plain log features instead: event-type mix, tool-output format and schema fields. These features find four undocumented changes. They are the answerer swap (04-20), a search-tool schema change (07-29), a two-day search malfunction and the end of the bookends (08-04). EP stays a descriptor of how directed the agents' tool workflows are. One post hoc lead needs a holdout check: at scaffold changes, agents' EP moves in the same direction more often than on random days.

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RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P0 synthetic calibration	false alarms 0.03–0.05; power met (fine) 0.78/0.89 at $\pm 50\%$ (fine), 0.13 (coarse)	
P1 scaffold changes jump	tool class $p = 0.23$, 0/17 hits; failed prompts 0/15	
P2 no jump: goal, roster, room	roster 0/21, room 1/5; goal 3/28 mixed ($p = 0.08$), agent-only $p = 0.007$	
P3 scaffold > goal	$\Delta t = -0.06$ ($p = 0.60$)	failed (R1)
P4 blind change-points near scaffold	$0.61\times$ chance ($p = 0.87$); no can-didate with $f_+ \geq 0.8$	failed
P6 family effect vanishes	η^2 0.34 $\rightarrow$ 0.40, $p = 0.0005$ both	failed (R4)
P7 NE43 no jump	$p \geq 0.38$, 3 designs (non-held diagnostic)	held
P8 NE14b jumps	8/12 agents fall, $t = -1.33$, $p = 0.33$	failed
P9 NE40 dated, EP sees it	dated 04-20 (post hoc stylometry); $t = -0.18$	failed
P10 #51 no roster/enrichment	$0.69\times$ ($p = 0.93$); trend $\rho = -0.40$	mixed

GOAL PERIODS IN DETAIL

G04 $\times$ failed
G05 $\times$ failed
G06 $\times$ failed
G07 $\times$ failed
G08 $\times$ failed
G10 $\times$ failed
G11 d descriptive
G12 $\times$ failed
G13 $\times$ failed
G16 $\times$ failed

+28 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

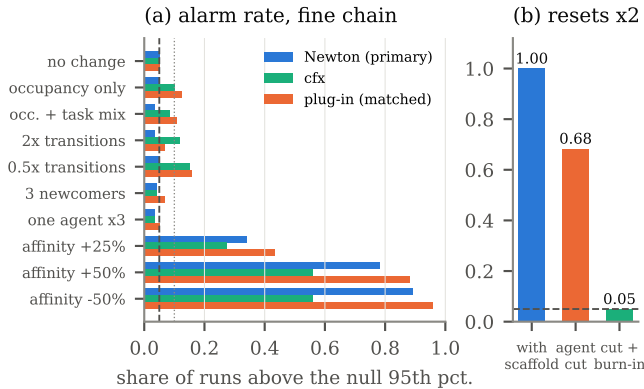
CAVEATS

Changes under about 30% of fine-chain EP are invisible, and coarse chains are blind; NE14b's -28% sits at the edge. The event catalog is dense and confounded: only 1 tool change, 2 prompt changes and 5 kickoffs have clean windows. Changelog dates are merge dates. Providers can respond in opposite directions (NE14b), and the mean-change t dilutes that. Amendment 3 repairs, after the data, a placebo pool too small to give $p < 0.11$; no verdict depends on it. NE40's date comes from post hoc stylometry. All values are lower bounds (non-Markov chains). The holdout is not run.

ROUND 2

- **Essence**. a log-only detector of platform changes the operator did not document.
- **H56-R1** Detect platform changes from direct log features (event mix, tool schemas, output stylometry), blind-scored against the changelog.
- **H56-R2** Cross-agent sign synchrony (the post hoc lead) as the platform signature: holdout C6, per-provider DiD.
- **H56-R3** EP on Jev v3 work states, split into task-driven and scaffold-driven parts.

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Second figure. Synthetic check at village sampling (12 agents, 3+3 days). (a) Share of runs above the null's 95th percentile. The cross-fitted Newton bound ignores occupancy, count, roster and single-agent changes and detects affinity changes. The plug-in and cross-fitted exact dual respond to counts. (b) A scaffold reset leaves a footprint on the cut agent-only chain; a 3-transition burn-in removes it.